

Chapter 19 Homework Answer Sheet

Name: _____ Student ID: _____ Instructor: _____ Score: _____

Part A multiple choice (18 points)

1-5: ACBAB

6-10: CBBCD

11-15: DACAB

16-18: CAD

Part B short question (20 points)

19 (4 points)

(a) (2 points) At $P=1$ atm, $T_m=0$ °C and ice cube and surroundings has infinitesimally small difference in temperature.

(b) (2 points) No. Melting is an endothermic process that increases molecules distance (entropy of the system). Even though there is no temperature change.

20 (6 points)

(a) (2 points)

$$\begin{aligned}\Delta H^\circ &= \Delta H_f^\circ[\text{CH}_3\text{OH}(g)] - 1/2\Delta H_f^\circ[\text{O}_2(g)] - \Delta H_f^\circ[\text{CH}_4(g)] \\ &= -201.2 - 1/2 \times 0 - (-74.8) = -126.4 \text{ kJ} \\ \Delta S^\circ &= S^\circ[\text{CH}_3\text{OH}(g)] - 1/2 S^\circ[\text{O}_2(g)] - S^\circ[\text{CH}_4(g)] \\ &= 237.6 - 1/2 \times 205 - 186.3 = -51.2 \text{ J/K}\end{aligned}$$

(b) (1 point)

Increase.

(c) (1 point)

$$\begin{aligned}\Delta G^\circ &= \Delta H^\circ - T\Delta S^\circ = -126.4 - 298 \times (-51.2)/1000 = -111.1 \text{ kJ} \\ \text{OR } \Delta G^\circ &= \Delta G_f^\circ[\text{CH}_3\text{OH}(g)] - 1/2 \Delta G_f^\circ[\text{O}_2(g)] - \Delta G_f^\circ[\text{CH}_4(g)] \\ &= -161.9 - 1/2 \times 0 - (-50.8) = -111.1 \text{ kJ}\end{aligned}$$

(d) (2 points)

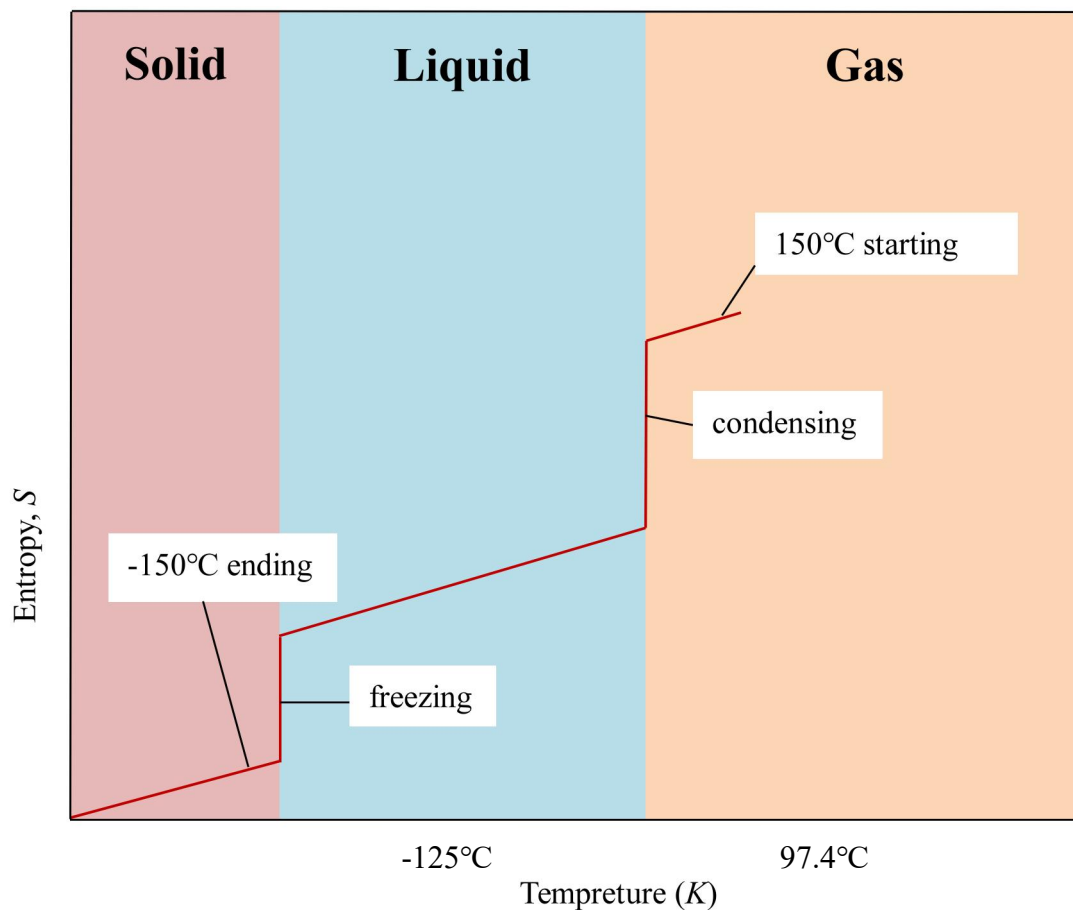
$$\begin{aligned}\Delta G &= \Delta H - T\Delta S = -126.4 - T \times (-51.2)/1000 \\ T &= 2468.75 \text{ K}\end{aligned}$$

No. This temperature is so high that the reactants and products are likely to decompose. At standard conditions, equilibrium is functionally unattainable for this reaction.

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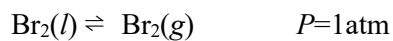
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21 (4 points)



22 (6 points)

(a) (2 points)



(b) (2 points)

at equilibrium $\Delta G^\circ = 0$

(c) (2 points)

$$\Delta H^\circ = 30.71 \text{ kJ}$$

$$\Delta S^\circ = 245.3 - 152.3 = 93.0 \text{ J/K}$$

$$\Delta G^\circ = \Delta H^\circ - T_b \Delta S^\circ = 0$$

$$T_b = \Delta H^\circ / \Delta S^\circ - 273 = 57.2^\circ\text{C} \quad \text{OR} \quad 330. \text{ K}$$